

TECHNICAL ASSESSMENT EMBEDDED INTERN

Water Management Controller
(IoT)

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INTRODUCTION

Water is one of the most critical resources in residential, agricultural, and industrial systems. Efficient monitoring and control of water distribution systems is essential to avoid wastage, prevent pump damage, and ensure continuous supply.

This project presents a **Water Management Controller using STM32 and ESP32 with Firebase IoT integration**, designed to enable real-time monitoring and remote control of a water distribution system.

The system uses an **STM32 microcontroller** as the main sensing and control unit. It collects data from various sensors such as tank level, sump level, flow rate, dry-run detection, and overflow conditions. Based on these inputs, it performs local decision-making for pump protection and safety.

The collected data is transmitted to an **ESP32 module via UART communication**, which acts as an IoT gateway. The ESP32 uploads this data to **Firebase Realtime Database**, enabling cloud-based monitoring through web or mobile applications.

In addition to monitoring, the system supports **remote control functionality**, where commands such as pump ON/OFF, valve control, and buzzer activation are received from Firebase and executed by the ESP32. These actions are also reflected using LED indicators for simulation purposes.

This dual-controller architecture ensures both **real-time embedded control (STM32)** and **cloud connectivity (ESP32 + Firebase)**, making the system suitable for smart water management applications in modern IoT environments.

OBJECTIVE

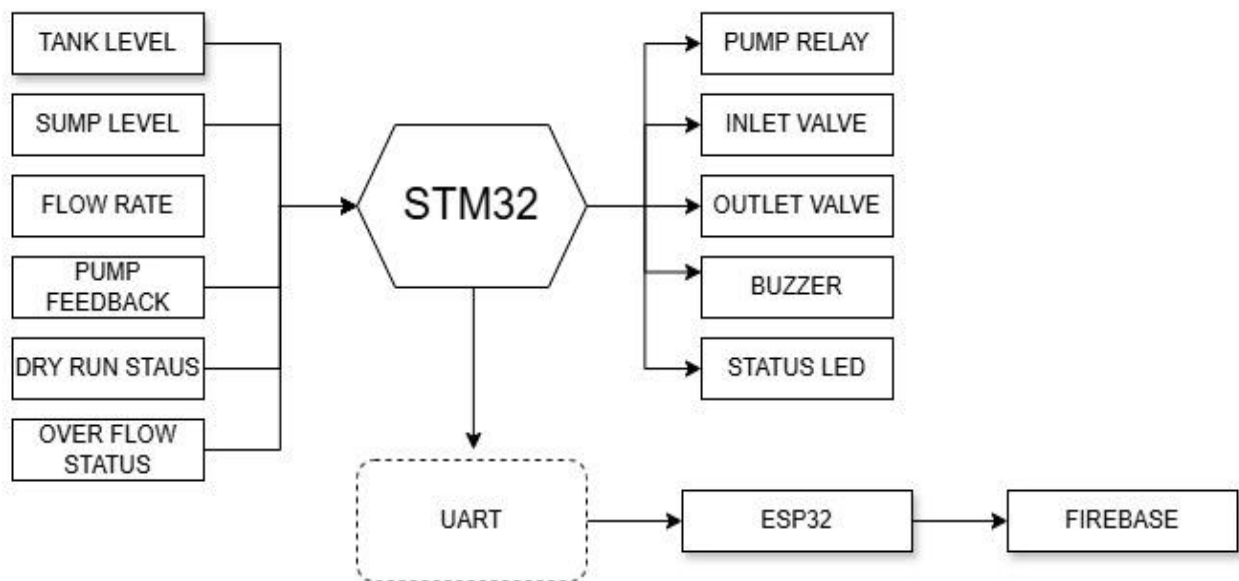
The main objective of this project is to design and develop a **smart water management system** using **STM32 microcontroller and ESP32 with Firebase IoT integration**, enabling real-time monitoring and control of water distribution processes.

The specific objectives of this project are:

- To monitor critical water parameters such as **tank level, sump level, flow rate, dry-run status, and overflow condition** in real time.
- To ensure safe operation of the pump by implementing **fault detection and protection logic** (dry-run and overflow protection).
- To control water distribution components such as **pump relay, inlet valve, outlet valve, and buzzer** based on system conditions.
- To transmit sensor data from **STM32 to ESP32 using UART communication** efficiently and reliably.
- To upload real-time sensor data to **Firestore Realtime Database** for remote monitoring.
- To enable **cloud-based control** by receiving commands from Firebase such as pump ON/OFF and valve control.
- To provide visual indication of system status using **LED-based simulation outputs**.
- To develop a scalable IoT-based architecture suitable for **smart irrigation and industrial water management applications**.

BLOCK DIAGRAM

WATER MANAGEMENT CONTROLLER
SYSTEM BLOCK DIAGRAM



WORKING PRINCIPLE

1. Sensor Data Acquisition (STM32)

The STM32 continuously reads data from connected sensors:

- Tank level sensor → measures main tank water level
- Sump level sensor → monitors sump availability
- Flow sensor → measures water flow rate
- Dry-run sensor → detects pump running without water
- Overflow sensor → detects tank overflow condition
- Pump feedback → confirms pump ON/OFF status

All sensor values are processed in real time.

2. Local Control & Safety Logic (STM32)

STM32 applies automated decision-making logic:

- If dry-run is detected → pump is immediately stopped
- If overflow occurs → pump is turned OFF and alarm is triggered
- Based on tank level → pump is automatically controlled
- Ensures safe and efficient water usage

Outputs controlled:

- Pump relay
- Inlet valve
- Outlet valve
- Buzzer
- Status LED

HARDWARE REQUIRED

1. Microcontrollers

- **STM32 (e.g., STM32F103C8T6 Blue Pill)**
Used for real-time sensor reading, control logic, and safety operations.
- **ESP8266 / ESP32 (NodeMCU preferred)**
Used as IoT gateway for Firebase communication and remote control.

2. Sensors

- **Water Level Sensor (Tank Level)** – measures main tank water level
- **Water Level Sensor (Sump Level)** – measures sump water availability
- **Flow Sensor (YF-S201 or similar)** – measures water flow rate
- **Dry Run Sensor** – detects pump running without water
- **Overflow Sensor** – detects excess water condition
- **Pump Feedback Signal** – confirms pump ON/OFF status

3. Actuators / Output Devices

- **Relay Module (5V/12V)** – for pump control
- **Inlet Valve (Solenoid Valve)** – controls water inlet
- **Outlet Valve (Solenoid Valve)** – controls water outlet
- **Buzzer** – alarm indication
- **LEDs (Status Indicators)** – pump, fault, valve status simulation

4. Power Supply

- **5V regulated power supply** for ESP32/ESP8266
- **3.3V supply (via regulator or onboard STM32 regulator)**
- Separate supply recommended for relays and pump

5. Communication Interface

- **UART (Serial Communication)** between STM32 and ESP32
- Optional level shifting (if required)

6. Connectivity Module

- **WiFi (built-in in ESP8266/ESP32)**
Used for Firebase cloud communication

7. Development Tools

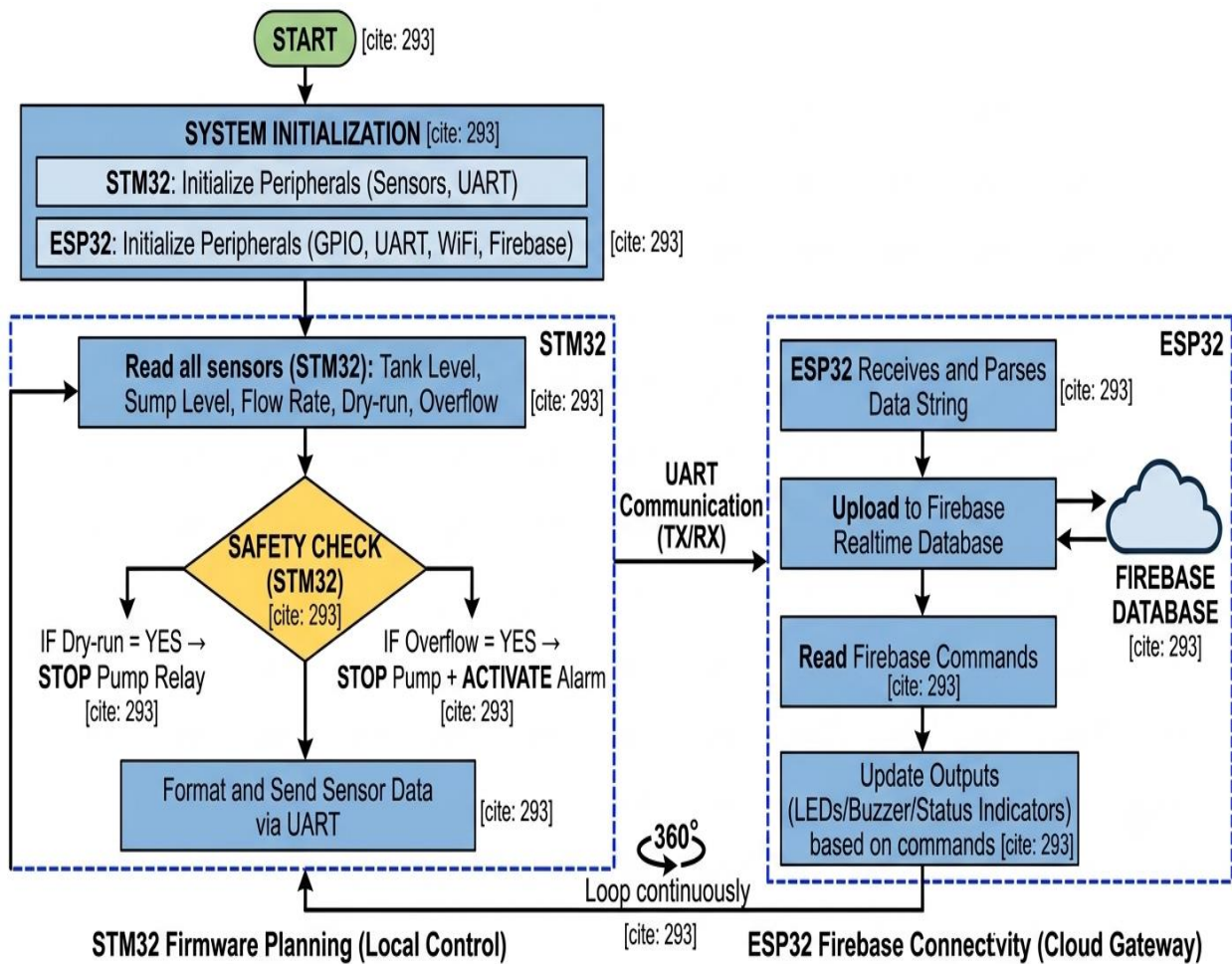
- USB cable for programming STM32 and ESP
- Breadboard / PCB for circuit assembly
- Jumper wires

SOFTWARE REQUIRED

1. IDE / Programming Tools

- Arduino IDE
Used for programming ESP8266 / ESP32 (NodeMCU) and uploading firmware.
- STM32CubeIDE (optional)
Used for STM32 firmware development, configuration, and debugging.

FLOW CHART



APPLICATION

The **Smart Water Management Controller using STM32 and ESP32 with Firebase integration** has wide applications in domestic, agricultural, and industrial environments where efficient water usage and monitoring are required.

1. Domestic Water Management

- Automated control of household water tanks
 - Prevents overflow and dry-run conditions
 - Remote monitoring using mobile/web applications
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2. Smart Irrigation Systems

- Used in agriculture for efficient water distribution
 - Automates irrigation based on tank and soil conditions (extendable)
 - Reduces water wastage and improves crop yield
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3. Industrial Water Control

- Monitors and controls water usage in factories
 - Protects pumps from dry-run and overload conditions
 - Ensures continuous and safe water supply
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4. Commercial Buildings

- Used in apartments, hotels, and offices
 - Centralized water tank monitoring system
 - Reduces manual supervision requirements
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5. IoT-Based Remote Monitoring

- Real-time data visualization using Firebase cloud

- Remote ON/OFF control of pumps and valves
- Alerts and fault notifications via cloud dashboard

ADVANTAGES

- **Real-time Monitoring:**
Continuously monitors tank level, sump level, flow rate, and fault conditions.
- **Automatic Pump Protection:**
Prevents pump damage using dry-run and overflow detection logic.
- **Remote Control via IoT:**
System can be controlled from anywhere using Firebase cloud.
- **Reduced Manual Effort:**
Eliminates need for manual switching of pumps and valves.
- **Efficient Water Usage:**
Optimizes water distribution and reduces wastage.
- **Dual Controller Architecture:**
STM32 handles real-time control while ESP32 handles cloud communication.
- **Scalable System:**
Can be easily expanded for industrial and agricultural applications.

DISADVANTAGES

- **Internet Dependency:**
Firebase features require stable internet connection.
- **Hardware Complexity:**
Uses multiple controllers (STM32 + ESP32), increasing system complexity.
- **Initial Setup Cost:**
Sensors, controllers, and relays increase initial cost.
- **Maintenance Requirement:**
Sensors may require periodic calibration and maintenance.

- **Latency Issue:**

Small delay may occur between cloud command and execution.

FUTURE IMPROVEMENTS

1. AI-Based Water Prediction System

- Use machine learning to predict water consumption patterns
- Automatically optimize pump scheduling based on usage history

2. Mobile Application Integration

- Dedicated Android/iOS app for monitoring and control

- Push notifications for alerts (overflow, dry-run, faults)
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3. Advanced Sensor Integration

- Add soil moisture sensors for smart irrigation
 - Add pressure sensors for pipeline monitoring
 - Improve accuracy using industrial-grade sensors
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4. Cloud Analytics Dashboard

- Real-time graphical dashboard using Firebase or other cloud platforms
 - Data analytics for water usage trends
 - Historical data storage and reporting
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5. Power Optimization

- Use low-power modes for STM32 and ESP32
- Solar-powered system for remote installations